

SIAM HOSSAIN

Backend & Full-Stack Engineer — Node.js · REST APIs · MongoDB/PostgreSQL | AI-Assisted Development

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SUMMARY

Backend-focused full-stack engineer with 2+ years building production software across Node.js/Express and Python — designing RESTful APIs, data models, and authentication for real, deployed applications. I ship AI-integrated features (OpenAI, RAG, agents) and use AI coding agents — Claude Code, Cursor, and GitHub Copilot — daily to build and review code, evaluating model output for correctness, edge cases, and maintainability before it ships. Strong CS fundamentals with 2,000+ algorithmic problems solved; effective working independently and asynchronously in remote teams.

EXPERIENCE

Freelance Software Engineer — Backend & Full-Stack

Remote · 2025 – Present

- Built and deployed a production **Pharmacy POS for Nasema Pharmacy** (running live on the shop counter): backend for billing, inventory, role-based access control, invoicing, and reporting on a Node.js + MongoDB stack.
- Designed RESTful APIs, MongoDB/SQL data models, and JWT authentication/authorization; owned features end-to-end — from data and APIs through to the UI — and deployed them.
- Shipped free web tools for local businesses — a Google Sheets-backed stock-audit app and a multi-business invoice generator — handling real business data behind access control.
- Use **Claude Code, Cursor, and GitHub Copilot** daily to implement and review code, validating and comparing model output for correctness, edge cases, and maintainability before merging.

Independent & Open-Source Projects

2023 – 2025

- Designed and shipped full-stack and AI projects (e-commerce, automation, developer tooling — see below) alongside a CS degree, building 2+ years of hands-on engineering experience.

PROJECTS

AIUB Notice Bot — Serverless Automation Pipeline

2026

Python 3.12, GitHub Actions, BeautifulSoup, GitHub Models (LLM), Telegram Bot API

- Serverless pipeline scheduled 4x/day on GitHub Actions: scrapes university notices, deduplicates against persisted state, AI-classifies into 8 categories and summarizes, then pushes to Telegram — at \$0/month.
- Engineered for reliability: idempotent runs, flood-guard, fail-safe state persistence committed back to the repo, and single-run concurrency to prevent overlap.

E-Commerce Platform — Full-Stack (MERN)

2024

React, Node.js, Express, MongoDB, Stripe, JWT

- Built RESTful backend APIs for catalog, cart, and order management; integrated Stripe with a secure checkout and **webhook-based order confirmation**.
- Implemented JWT authentication and an admin dashboard for inventory, fulfillment, and sales analytics.

LLM Agents + RAG — AI Backend

2025

Python, FastAPI, OpenAI (function calling), FAISS / vector embeddings, LangChain

- Built a FastAPI service exposing OpenAI function-calling agents with tool use and memory, plus a RAG pipeline over documents using vector retrieval for grounded, source-backed answers.

TECHNICAL SKILLS

Backend: Node.js, Express.js, Python, FastAPI, RESTful API design, Authentication & Authorization (JWT), Webhooks

Databases: MongoDB, PostgreSQL, MySQL, Prisma ORM, schema & data modeling

AI & AI-Assisted Dev: Claude Code, Cursor, GitHub Copilot (daily) · OpenAI API, RAG, AI agents, prompt engineering · model-output review & evaluation

Languages: JavaScript (ES6+), TypeScript, Python, SQL

Frontend: React, Angular, TypeScript, Tailwind CSS (end-to-end delivery)

DevOps & Tools: Git, GitHub Actions (CI/CD), Docker, AWS, Linux, Postman

Practices: Agile/Scrum, code review, debugging, performance optimization, remote/async collaboration

EDUCATION

BSc in Computer Science & Engineering — American International University-Bangladesh (AIUB)

Graduated 2026

CERTIFICATIONS & ACHIEVEMENTS

- **Certifications:** Meta Front-End Developer (Coursera) · Google Generative AI Fundamentals · HackerRank Problem Solving & SQL
- **Achievements:** 2,000+ algorithmic problems solved across LeetCode, Codeforces & HackerRank · Active open-source contributor on GitHub
- **Languages:** Bangla (Native) · English (Professional) · German (Elementary)